

Introduction to Time-Varying Treatments

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- Focused on point exposure studies - wanted to evaluate the effect of a treatment given at a single time point or a single treatment decision made at a single time point on a response
- In many cases, however, treatment decisions are more complex, evolving over time
- Certainly the case for many chronic diseases and long-term conditions
- For example, cancer, HIV, cardiovascular disease, diabetes, hypertension, substance abuse, smoking, ADHD, mental health conditions

- Assume that treatment decisions are made at a finite number of decision points denoted by $t_0, \dots, t_M$.
- May be fixed or random times
- Examples: dose adjustment for hypertensive patients, whether or not to “start” treatment - e.g., when to do bariatric surgery, when to implant device, etc.

JAMA | **Original Investigation**

Association of Metabolic Surgery With Major Adverse Cardiovascular Outcomes in Patients With Type 2 Diabetes and Obesity

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Figure 1: JAMA; September 2, 2019

A Recent Example

DESIGN, SETTING, AND PARTICIPANTS Of 287 438 adult patients with diabetes in the Cleveland Clinic Health System in the United States between 1998 and 2017, 2287 patients underwent metabolic surgery. In this retrospective cohort study, these patients were matched 1:5 to nonsurgical patients with diabetes and obesity (body mass index [BMI] ≥ 30), resulting in 11 435 control patients, with follow-up through December 2018.

EXPOSURES Metabolic gastrointestinal surgical procedures vs usual care for type 2 diabetes and obesity.

MAIN OUTCOMES AND MEASURES The primary outcome was the incidence of extended MACE (composite of 6 outcomes), defined as first occurrence of all-cause mortality, coronary artery events, cerebrovascular events, heart failure, nephropathy, and atrial fibrillation. Secondary end points included 3-component MACE (myocardial infarction, ischemic stroke, and mortality) and the 6 individual components of the primary end point.

RESULTS Among the 13 722 study participants, the distribution of baseline covariates was balanced between the surgical group and the nonsurgical group, including female sex (65.5% vs 64.2%), median age (52.5 vs 54.8 years), BMI (45.1 vs 42.6), and glycated hemoglobin level (7.1% vs 7.1%). The overall median follow-up duration was 3.9 years (interquartile range, 1.9-6.1 years). At the end of the study period, 385 patients in the surgical group and 3243 patients in the nonsurgical group experienced a primary end point (cumulative incidence at 8-years, 30.8% [95% CI, 27.6%-34.0%] in the surgical group and 47.7% [95% CI, 46.1%-49.2%] in the nonsurgical group [$P < .001$]; absolute 8-year risk difference [ARD], 16.9% [95% CI, 13.1%-20.4%]; adjusted hazard ratio [HR], 0.61 [95% CI, 0.55-0.69]). All 7 prespecified secondary outcomes showed statistically significant differences in favor of metabolic surgery, including mortality. All-cause mortality occurred in 112 patients in the

A Recent Example

- Studies in which treatment is time-varying often “shoehorned” into looking like point-exposure studies
- What are the potential biases from doing that in this study?

Why Don't "Standard" Methods Work

- Obviously, this approach to estimating the mean response if all individuals followed a particular treatment assignment involves quite a bit of modeling
- Why can't we just specify one model for the outcome? Key followup questions - which covariates to include?
- If include just baseline covariates and treatment history then there will be residual confounding (biased treatment estimates)
- If include all the covariate history and treatment history then this will "block" some of the effect of treatment on outcome (that is some of the effect of early treatment on the outcome is through the treatment's effect on the longitudinal covariates)
- The later would essentially just estimate a controlled direct effect (to use terminology from mediation)

Observed Data

- Observed data from such longitudinal studies can be summarized as the time ordered variables
- $(L_0, A_0, L_1, A_1, \dots, L_M, A_M, Y)$
- L_0 - all the covariate information obtained prior to any treatment at decision point t_0 (baseline covariates)
- A_0 - first treatment given at time t_0 among a possible set of treatments $\mathcal{A}_0$
- L_j - denotes any additional covariates obtained between decision points t_{j-1} and t_j
- A_j - treatment given at the decision point t_j among the set of treatment $\mathcal{A}_j$
- Y - outcome of interest, response variable. Note that this is determined after the final treatment time t_M .

Overbar Notation

- Use the “overbar” notation to denote the history of the time-dependent variables. That is.,
- $\bar{L}_j = (L_0, \dots, L_j)$ and $\bar{A}_j = (A_0, \dots, A_j)$
- For brevity $\bar{L} = (L_0, \dots, L_M)$ and $\bar{A} = (A_0, \dots, A_M)$
- Thus the observed longitudinal data on subject i is given by $O_i = (\bar{L}_i, \bar{A}_i, Y_i)$

Potential Outcomes

- Set of potential outcomes given by
- $W = (L_0, L_1^{a_0}, L_2^{\bar{a}_1}, \dots, L_M^{\bar{a}_{M-1}}, Y^{\bar{a}_M})$
- for any $a_j \in \mathcal{A}_j$ for $j = 0, \dots, M$
- Note that we need to consider potential outcomes for both the “outcome” and all the time-dependent covariates

Key Question

- Can we deduce the distribution of the potential outcomes for a fixed $\bar{a}$ - that is the distribution of $(L_0, L_1^{a_0}, L_2^{\bar{a}_1}, \dots, L_M^{\bar{a}_{M-1}}, Y^{\bar{a}_M})$
- Based on the distribution of the observed data - that is the distribution of $(L_0, A_0, \dots, L_M, A_M, Y)$
- What assumptions will we need? Can we generalize consistency and no unmeasured confounding?

Key Assumptions

- ① Consistency: $Y = \sum_{\bar{a}_M \in \bar{\mathcal{A}}_M} Y^{\bar{a}_M} I(\bar{A}_M = \bar{a}_M)$
- ② Sequential Ignorability: A_j is independent of all potential outcomes (including $Y^{\bar{a}_M}$) given $\bar{L}_j$ and $\bar{A}_{j-1}$
- ③ Positivity: $0 \leq P(A_j = 1 | \bar{L}_j, \bar{A}_{j-1}) \leq 1$

Hypothetical Example

- Completely hypothetical example involving HIV disease circa 1990
- Patients come into the clinic every three months to have their CD4 counts measured
- Physicians have to decide whether or not to treat the patient for the next six months until their next visit
- The goal is to have a large a CD4 count as possible after two years
- Because it is known that treatment can eventually lead to mutation of the HIV virus and resistance testing, resistance testing is also conducted after the initiation of treatment

Working Assumptions

- log CD4 counts are (conditionally) normally distributed
- If treatment is effective then CD4 counts should increase from their previous value following treatment. However, we expect greater increases when CD4 counts are lower (i.e., an interaction between treatment and CD4 counts)
- Without treatment CD4 counts will decline
- The accumulation of treatment over time makes it more likely that a patient will become resistant to treatment
- Once a patient becomes resistant then their CD4 counts will decline with or without treatment

- t_0 is the first time a patient is seen; $t_j = 6j$ months afterward and $M = 3$. Response of interest is ascertained at 24 months
- For $j = 0, \dots, M$ let A_j denote the indicator of treatment assignment at time t_j
- Let $ICD4_j, j = 0, \dots, M + 1$ denote the log CD4 at time t_j
- Let $R_j, j = 1, \dots, M + 1$ denote the resistance status at time t_j where $R_j = 1$ if resistant and 0 otherwise

- Overall we have

$$L_0 = ICD4_0$$

$$L_1 = \{ICD4_1, R_1\}$$

$$\vdots$$

$$L_M = \{ICD4_M, R_M\}$$

$$Y = \{ICD4_{M+1}, R_{M+1}\}$$

Example Data

##	ID	R0	CD40	A0	R1	CD41	A1	R2	CD42	A2	R3	CD43	A3	R4	CD44
## 1	1	0	537	0	0	337	1	1	176	0	1	144	0	1	95
## 2	2	0	1116	0	0	825	1	0	558	1	0	671	0	0	645
## 3	3	0	632	0	0	592	0	0	897	0	0	899	0	0	605
## 4	4	0	722	0	0	321	1	0	517	1	0	812	1	1	798
## 5	5	0	401	0	0	364	1	0	321	0	0	295	0	0	321
## 6	6	0	446	1	0	487	0	0	401	1	0	745	0	0	1352